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Comparison of travel behaviour of two individuals with different tracking methods

PTED Semester Project

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1. Abstract

2. Introduction

This project investigates the movement behavior of two individuals. These two individuals did a tracking of their movement using different tracking methods, one using GPS-based app (Christopher) and one using Google Timeline [1] (Amelia). It applies descriptive spatio-temporal trajectory analysis to compare mobility behaviour recorded with two different tracking methods. The analysis focuses on movement trajectories as sequences of timestamped geographic coordinates. Further spatial analysis methods will include distance calculations, home-based radius analysis, and spatial overlay with OpenStreetMap [2] road network data to conclude frequently used road types. The interpretation of the results are supported by the participants' own knowledge of their daily movements.

The following research questions get analysed:

- Which road types are most frequently used based on the recorded trajectories?
- How far is the typical or maximal movement radius from their home when they travel?
- and how far do they travel on average?
- what is the average speed the individuals are travelling?
- How long are the individuals out of their home on average?
- How long does the travel time from home to ZHAW take based on the movement data?

As there are differences in data structure, sampling intervals, GPS accuracy and preprocessing effort in both datasets, an additional objective is to evaluate how different tracking methods influence the quality and reliability of Trajectories, movement analysis and processability.

3. Material and Methods

3.1. Data

This project uses two primary mobility data sources, tracked from two individuals: Google Timeline data and GPS Logger data. Google Timeline provides sparse but semantically enriched location records, including latitude/longitude coordinates, timestamps, and partially pre-classified activity labels (e.g., walking, driving, cycling). However, its temporal resolution is low and irregular, which limits fine-grained movement analysis. The Coordinates are not as precise as GPS data and can deviate several meters from the known location. Over a tracking period of 17 days (08.04.2026 to 24.04.2026), a total of 5′420 points were recorded, of which 658 points were attributable to movement-related activity.

In contrast, GPS Logger data offers high-frequency, continuous tracking at fixed time intervals, containing raw coordinates and timestamps. This allows for detailed trajectory reconstruction but introduces noise, tracking holes (crashing app) and requires preprocessing.

3.2. Used R packages

The project is implemented in R [3], [4] using a combination of data wrangling, time-series handling, spatial analysis, and cartographic visualisation packages. The tidyverse and lubridate packages get used for cleaning, restructuring, filtering, summarising the movement data and management of time data. Spatial processing is mainly carried out with the sf package. OpenStreetMap road network data is accessed using osmdata and linked to the mobility trajectories through nearest-feature matching. For visualisation, ggplot2 gets used for statistical plots and the package tmap will be used to display the movement data on maps. Matched road types gets displayed using leaflet. Other packages used for processing or plotting include; jsonlite, dplyr, purrr, stringr, osmextract, data.table

3.3. Pre processing

google timeline data

1. Importing Google Timeline JSON data, extracting coordinates, timestamps and convert them, as well as delete errors.
2. Calculating distances and time differences between two consecutive points. From these, the moving window is computed as the mean of both neighboring points.
3. Detecting stationary and moving phases using distance-based thresholds (< 100 m).
4. Estimating relevant locations such as home and ZHAW with most frequent coordinates in certain time frames.

GPS data:

3.4. Parameter calculation

1. Preparing the use of OpenStreetMap and its categories which includes loading the Swiss road/rail network lines, filtering to relevant categories, grouping these categories and match the points to the nearest feature.
2. Calculating mobility indicators:

- daily travelled distance : sum of distance per day.

- average movement speed: computed from distance/time.
- time spent outside home: sum of time per day.
- maximum daily movement radius: maximum distance to home per day.
- travel time from home to ZHAW: arrival/departure times at university and home.
- road type use: calculation of time shares per transport group per day and across the entire tracking period. Additional for Google Timeline data: cross-tabulation/heatmap comparing Google Activity labels with OSM transport group.

3. Visualising trajectories and movement patterns using maps and plots.

3.5. Validation

The validation of the different calculated parameters and the road type matching with OpenStreetMap is based on methods described by [5]. The used methods include visualization and event validity. As the tracking data was collected by the authors, they do have knowledge of the correct events and can assign errors and estimate miscalculated values.

4. Results

4.1. Google timeline data

After moderate pre processing effort, the google timeline data over more than two weeks looks like in Figure 1. Blue are the movement points and red are the static points. As some of the trajectories have little data points, the travelled distances and speed are likely to be underestimated.

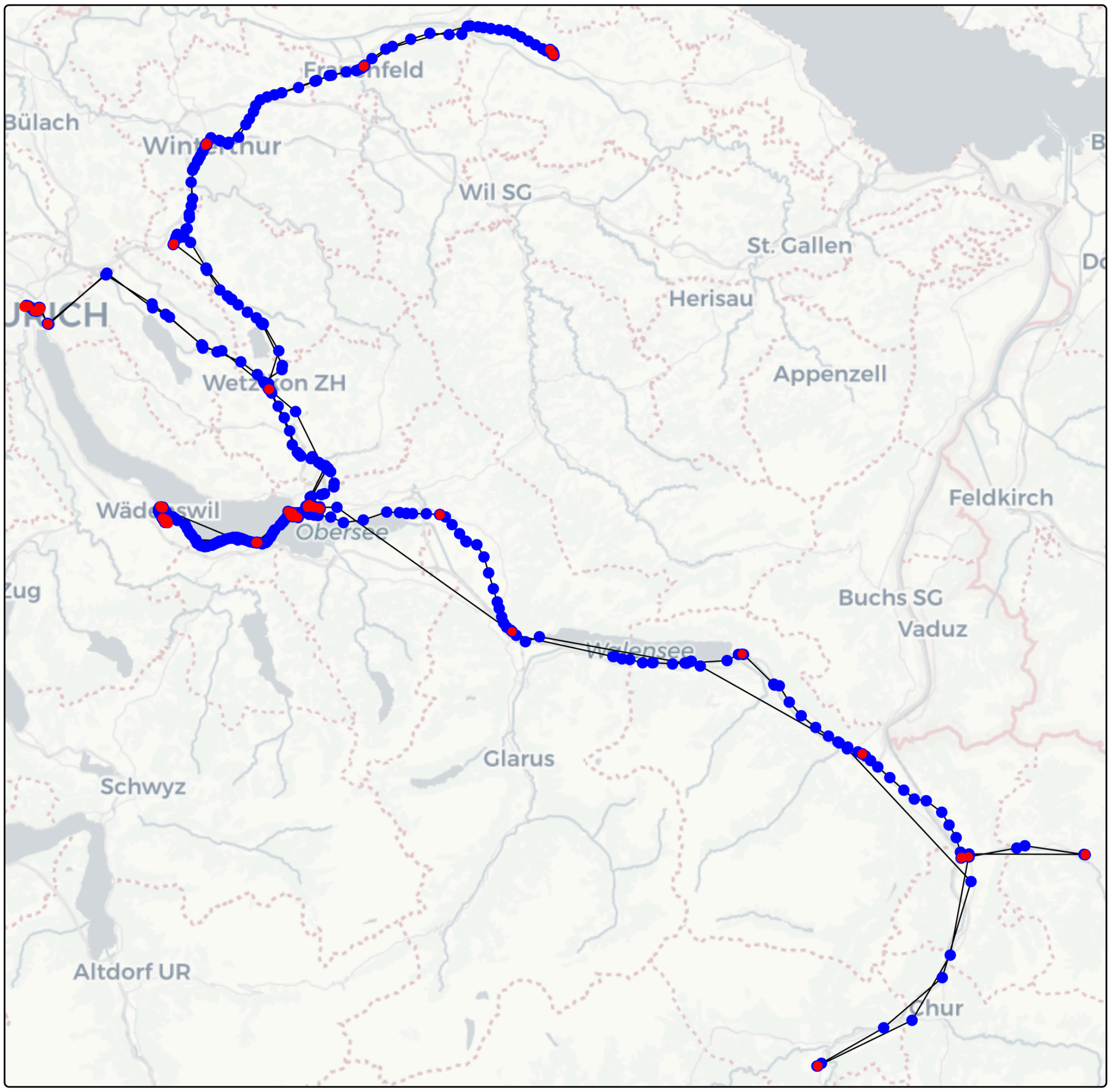

Figure 1: Tracked moving data with google timeline, red is static and blue is moving.

In the google timeline data, the accuracy is stated to be up to several hundred meters. Therefore the matching with roadtypes can be difficult, as roads are only few meters wide. An example of road type mismatch is the way from home to ZHAW and back in Figure 2. From Rapperswil train station to Wädenswil train station, only the train was used. But in the road type matching, several points are not matched as 'Train' and sometimes even far away from the railroads.

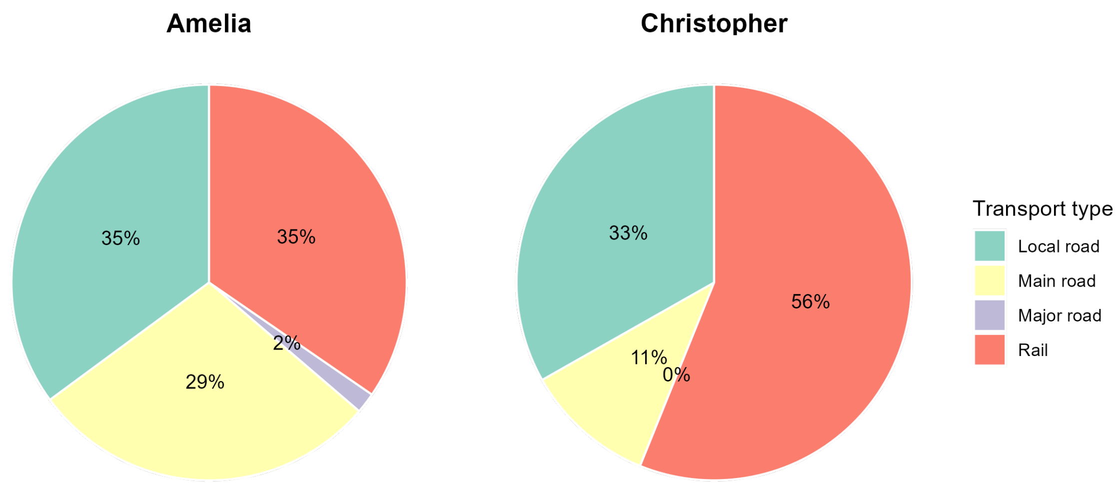

Figure 2: Example of mismatched road types on the way from home to ZHAW (only train from Rapperswil to Wädenswil).

Another attempt to evaluate the accuracy of the road type matching is to have a look at the from google stated activities at these points. Of course these activities can be wrong as well, but as they include more than the coordinates of the points (acceleration etc.) in their analysis, it is estimated to be more accurate. The percentages of different road types within the different google activities is shown in Figure 3.

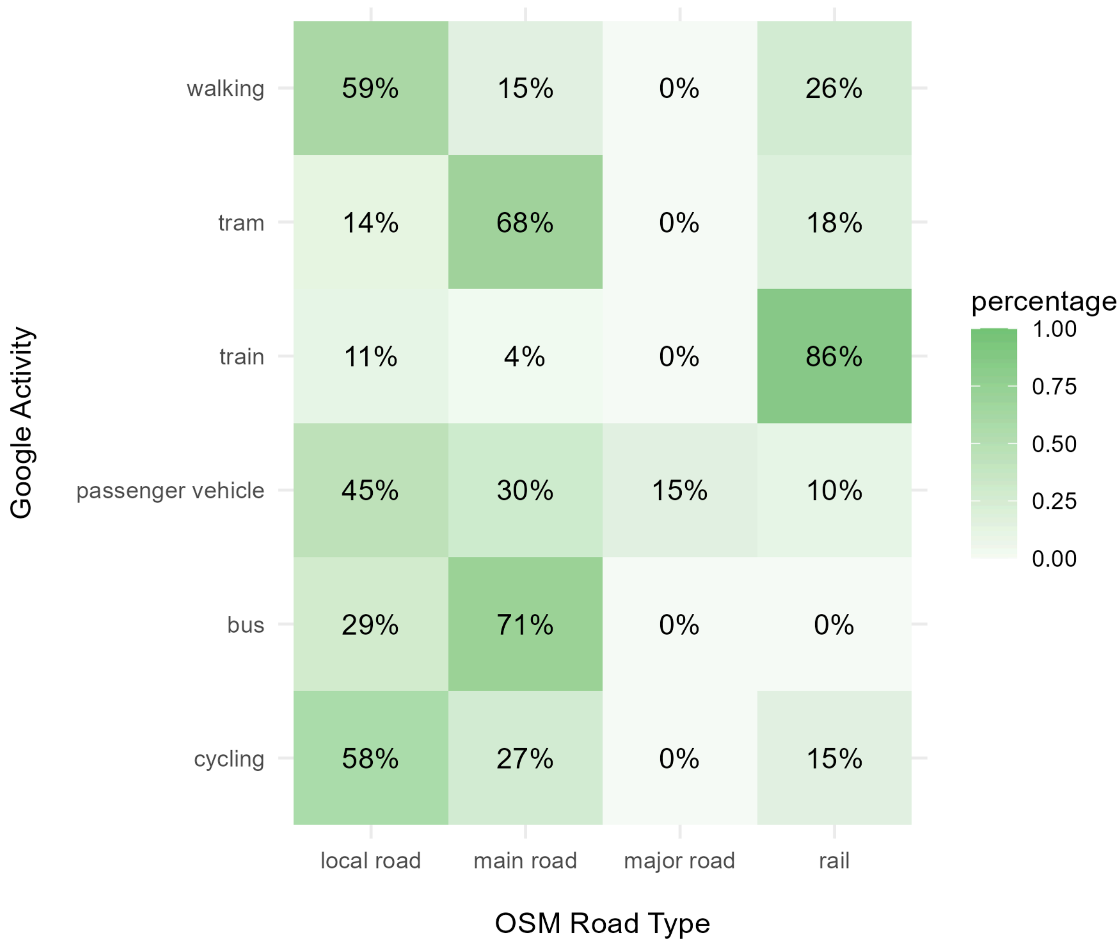

Figure 3: Percentage of different road types within the google matched activity.

At a first glance, some of the road types seem to be matched very wrong, but most of them can somehow be explained. The 26% of walking on the rail might be the changing of trains at train stations, the tram line is inside of the main road and therefore 68% matched with it. Nevertheless, a lot of the 4% to 15% (e.g. cycling on rail or train on local road) cannot be explained, other than errors.

The other parameters are calculated once per day and once an average over the whole tracking period. In Figure 4 it is shown, that the parameters vary widely at different days.

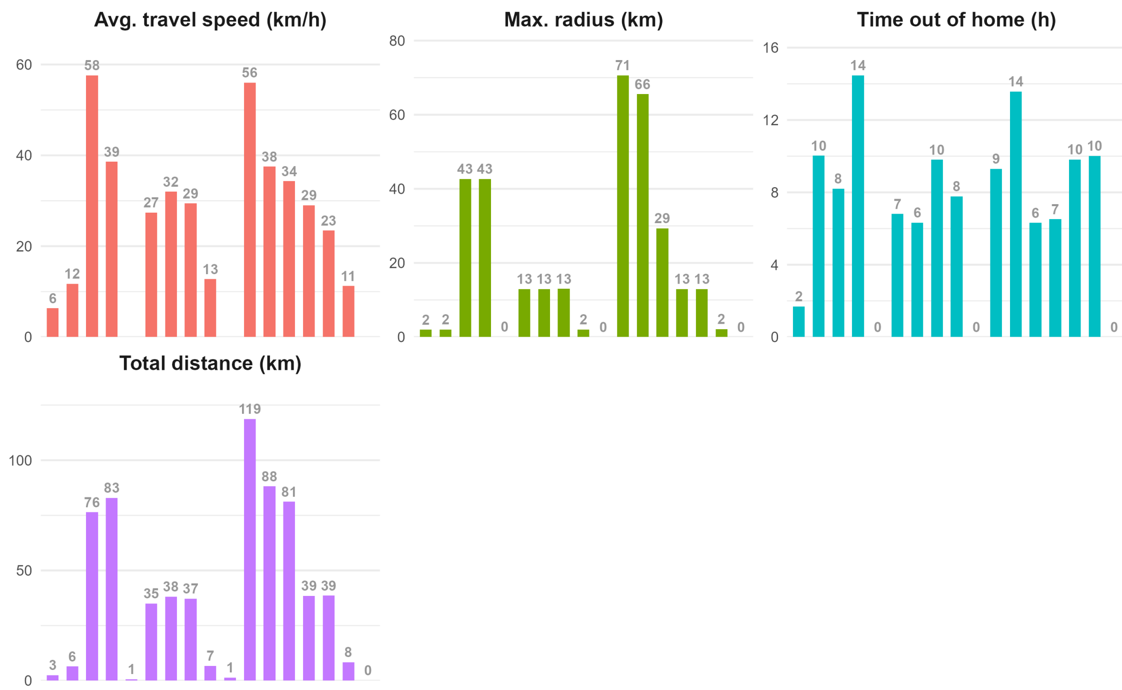

Figure 4: Parameter results of average travel speed, maximal distance from home, time out of home and total travelled distance for every tracked day.

4.2. GPS data

4.3. Comparison

5. Discussion

5.1. comparability

5.2. problems, inaccuracies, processing efforts

Cleaning, filtering, synchronizing timestamps, and handling missing values are expected to be time-consuming tasks. Furthermore, differences in sampling intervals between the two datasets may influence calculated metrics such as average speed or daily travelled distance, making direct comparisons challenging. Additionally, the positional uncertainty may complicate analyses such as road type assignment, where small spatial inaccuracies can incorrectly match trajectories to nearby streets or paths.

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Appendices

A. Appendix

A.1. Usage of AI Models

Complementary, the following AI was used for the listed areas in this project: [6] [7]

- Paraphrasing text
- support with coding for more complex processing

The authors of this report take the responsibility for the content. They declare to check the relevance, truthfulness and accuracy of the generative AI systems used and the output they produce and to use it in a reflective, critical and targeted manner. They also declare to not upload personal data and other data requiring heightened protection as well as data that is subject to contractual or legal confidentiality. [8]

A.2. Statement of authorship

The students affirm independent completion of their work without outside help. The students declare that all printed and electronic sources used in the text and the bibliography are correctly indicated, i.e., that the work does not contain any plagiarism. In the event of misconduct of any kind, Paragraph 39 and Paragraph 40 of the General Academic Regulations for Bachelor's and Master's degree programmes at the Zurich University of Applied Sciences (dated 29 January 2008) and the provisions of the Disciplinary Measures of the University Regulations shall apply. [9]

A.3. Wordcount

```
Error in `loadNamespace()`:
! es gibt kein Paket namens 'wordcountaddin'
```